

```
"""
```

```
CaptionBot - voice note (or typed text) -> WhatsApp status caption.  
Runs 100% on your laptop with Ollama + Gemma. No pip installs needed.
```

1. Make sure Ollama is running and you have pulled the model: `ollama pull gemma3`
2. Run: `python captionbot.py`
3. Open `http://localhost:8000` in CHROME (voice needs Chrome or Edge)

```
"""
```

```
import json, os, re, urllib.request  
from http.server import BaseHTTPRequestHandler, HTTPServer
```

```
MODEL = os.environ.get("MODEL", "gemma3:1b")  
OLLAMA = "http://localhost:11434/api/generate"
```

```
# Swap these for your seller's real listings later. Her own style = better captio  
EXAMPLES = """Details: 3 senator wear, sizes M L XL, 25k, Abuja delivery  
Caption: Fresh senator wear just landed 🔥 Sizes M, L, XL. 25k each. Abuja delive
```

```
Details: 2 ankara skirt, size 8 and 10, Port Harcourt delivery  
Caption: Two ankara skirts, sizes 8 and 10 ✨ Port Harcourt delivery available. D
```

```
def build_prompt(details, strict=False):  
    extra = "You MUST include the price exactly as given. " if strict else "  
    return f"""You write WhatsApp status captions for a Nigerian clothing seller.  
The details may be in Nigerian Pidgin. Understand it, but write the caption in si  
Pidgin help: "I get" = I have. "na" = is. "dey" = is/are. "fit deliver" = can del  
Rules: Write ONE caption only. No options, no notes, no questions.  
Use ONLY the details given. Never invent discounts, gifts, deadlines or style wor  
If no price is given, do not mention any price or amount.  
{extra}Keep the item names, sizes, price and delivery exactly as given.  
No placeholders. End with "DM to order". Under 40 words, max 3 emojis.
```

```
{EXAMPLES}
```

```
Details: {details}  
Caption:"""
```

```
def ask_ollama(prompt):  
    body = json.dumps({  
        "model": MODEL, "prompt": prompt, "stream": False,  
        "options": {"temperature": 0.3, "num_predict": 120},  
    }).encode()  
    req = urllib.request.Request(OLLAMA, body, {"Content-Type": "application/json"  
    with urllib.request.urlopen(req, timeout=120) as r:
```

```

        text = json.loads(r.read())["response"].strip()
    text = text.split("\n\n")[0].strip().strip(' ').strip()
    return text

def facts(details):
    """Prices/numbers from the details that the caption must keep, e.g. 18k, 4000
    return re.findall(r"\d[\d,.*]*\s?k?", details.lower())

def normalize(details):
    """Fix common mishears: '12 key' / '12 kay' / '12 thousand' -> '12k'."""
    d = re.sub(r"(\d)\s*(key|kay|ki|k\.|thousand|grand)\b", r"\1k", details, flag
    d = re.sub(r"\blist (blouse|gown|top|dress)", r"lace \1", d, flags=re.I)
    return d

def generate(details):
    details = normalize(details)
    need = [f.replace(" ", "") for f in facts(details)]
    best = ""
    given_prices = {p.replace(" ", "") for p in re.findall(r"\d[\d,.*]*\s?k\b", de
    for attempt in range(3):
        cap = ask_ollama(build_prompt(details, strict=attempt > 0))
        flat = cap.lower().replace(" ", "")
        # reject captions that invent a price that was never given
        invented = [p for p in re.findall(r"\d[\d,.*]*k\b", flat) if p not in give
        if all(n in flat for n in need) and not invented:
            return cap, attempt + 1
        best = best or cap
    return best, 3

PAGE = r"""<!doctype html>
<html lang="en"><head><meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>CaptionBot</title>
<style>
:root{--indigo:#1b2a6b;--saffron:#f2a900;--cream:#fbf7ef;--ink:#1a1a1a;--green:#1
*{box-sizing:border-box}
body{margin:0;background:var(--cream);color:var(--ink);font-family:"Trebuchet MS"
header{background:var(--indigo);color:#fff;padding:28px 20px 22px;border-bottom:6
header h1{margin:0;font-size:1.7rem;letter-spacing:.2px}
header p{margin:6px 0 0;opacity:.85;max-width:44ch}
main{max-width:560px;margin:0 auto;padding:20px}
label{display:block;font-weight:bold;margin:0 0 6px}
textarea{width:100%;min-height:110px;padding:12px;border:2px solid var(--indigo);
button{font:inherit;font-weight:bold;border:0;border-radius:6px;padding:12px 16px
button:focus-visible,textarea:focus-visible{outline:3px solid var(--saffron);outl
.row{display:flex;gap:10px;margin:12px 0;flex-wrap:wrap}
#mic{background:var(--saffron);color:var(--ink);flex:1}

```

```

#mic.on{background:#c0392b;color:#fff}
#go{background:var(--indigo);color:#fff;flex:1}
button:disabled{opacity:.5;cursor:wait}
#out{display:none;margin-top:18px;background:#fff;border-left:6px solid var(--saf
#cap{white-space:pre-wrap;font-size:1.05rem;margin:0 0 12px}
#copy{background:#e8e8e8}
#wa{background:var(--green);color:#fff}
small{color:#555;display:block;margin-top:10px}
#err{color:#c0392b;margin-top:10px}
</style></head><body>
<header><h1>CaptionBot</h1><p>Say what you are selling. Get a WhatsApp status cap
<main>
<label for="d">What are you selling?</label>
<textarea id="d" placeholder="2 ankara gown, size 12 and 14, 18k each, Lagos deli
<div class="row">
<button id="mic" type="button">Tap to speak</button>
<button id="go" type="button">Write caption</button>
</div>
<div id="err" role="alert"></div>
<div id="out">
<p id="cap"></p>
<div class="row"><button id="copy" type="button">Copy</button><button id="wa" typ
<small id="meta"></small>
</div>
</main>
<script>
const d=document.getElementById('d'),mic=document.getElementById('mic'),go=docume
out=document.getElementById('out'),cap=document.getElementById('cap'),err=documen
const SR=window.SpeechRecognition||window.webkitSpeechRecognition;
let rec=null,listening=false;
if(!SR){mic.disabled=true;mic.textContent="Voice needs Chrome or Edge"}
else{
  rec=new SR();rec.lang='en-NG';rec.interimResults=true;rec.continuous=true;
  let base='';
  rec.onresult=e=>{let t='';for(let i=0;i<e.results.length;i++)t+=e.results[i][0]
  rec.onend=()=>{listening=false;mic.classList.remove('on');mic.textContent='Tap
  rec.onerror=e=>{err.textContent='Mic problem: '+e.error+'. Allow the microphone
  mic.onclick=()=>{
    err.textContent='';
    if(listening){rec.stop();return}
    base=d.value;listening=true;mic.classList.add('on');mic.textContent='Listenin
  };
}
go.onclick=async()=>{
  const details=d.value.trim();err.textContent='';
  if(!details){err.textContent='Say or type what you are selling first.';return}
  if(listening&&rec)rec.stop();

```

```

go.disabled=true;go.textContent='Writing...';out.style.display='none';
try{
  const r=await fetch('/generate',{method:'POST',headers:{'Content-Type':'appli
  const j=await r.json();
  if(!r.ok)throw new Error(j.error||'Failed');
  cap.textContent=j.caption;out.style.display='block';
  meta.textContent=j.tries>1?'Took '+j.tries+' tries to keep your price and siz
}catch(e){err.textContent='Could not write the caption. Is Ollama running? ('+e
go.disabled=false;go.textContent='Write caption';
};
document.getElementById('copy').onclick=async e=>{
  try{await navigator.clipboard.writeText(cap.textContent);e.target.textContent='
  setTimeout(()=>e.target.textContent='Copy',1800);
};
document.getElementById('wa').onclick={()=>window.open('https://wa.me/?text='+enco
</script></body></html>""

```

```

class H(BaseHTTPRequestHandler):
    def _send(self, code, body, ctype="application/json"):
        self.send_response(code)
        self.send_header("Content-Type", ctype + "; charset=utf-8")
        self.end_headers()
        self.wfile.write(body if isinstance(body, bytes) else body.encode())

    def do_GET(self):
        self._send(200, PAGE, "text/html")

    def do_POST(self):
        if self.path != "/generate":
            return self._send(404, json.dumps({"error": "not found"}))
        try:
            n = int(self.headers.get("Content-Length", 0))
            details = json.loads(self.rfile.read(n))["details"][:500]
            caption, tries = generate(details)
            self._send(200, json.dumps({"caption": caption, "tries": tries}))
        except Exception as e:
            self._send(500, json.dumps({"error": str(e)}))

    def log_message(self, *a):
        pass

if __name__ == "__main__":
    print(f"CaptionBot running with {MODEL}. Open http://localhost:8000 in Chrome
    HTTPServer(("127.0.0.1", 8000), H).serve_forever()

```